BITS Pilani - Apex Project

BS Program | Trimester 3

Project Report

P4 - Retail and E-Commerce: Conversational Inventory AI Pipeline

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Section 1: Problem Background

Retail and e-commerce businesses generate large volumes of transaction data every day, across products, customers, payment methods, and store locations. But raw data is rarely clean. Missing values, wrong data types, and inconsistent formatting are common, and if left unaddressed, they quietly undermine any analysis built on top of them. For a business relying on this data to make inventory or sales decisions, that is a real problem.

The dataset for this project comes from Kaggle and captures 12,575 retail transactions across 11 attributes, covering everything from product details and pricing to payment method, location, and discount status. On first inspection, it had several quality issues: 1,213 rows had no item name recorded, 609 were missing a price, 604 were missing quantity and total spend figures, and over 4,000 rows had no discount information at all. On top of that, transaction dates were stored as plain text, quantities as decimals instead of whole numbers, and discount values as text strings instead of proper True/False booleans.

The goal of this project was to build a step-by-step data cleaning pipeline to fix these issues, removing rows where the missing data could not be recovered, filling gaps where a reasonable estimate was possible, correcting data types, standardizing text fields, and preparing the dataset for analysis. The final output is a cleaned dataset of 11,362 rows, a before/after quality report, and three charts showing patterns in the cleaned data.

This project does not build any machine learning model, deploy any system, or make business recommendations. It is scoped entirely to data preparation and basic exploratory analysis.

Section 2: Dataset Description

Attribute	Details
Dataset Name	Retail Store Sales - Dirty for Data Cleaning
Source	Kaggle https://www.kaggle.com/datasets/ahmedmohamed2003/retail-store-sales-dirty-for-data-cleaning
Size	Raw: 12,575 Rows x 11 Columns Clean: 11,362 Rows x 17 Columns
Format	CSV (comma-separated values)
Key Fields	Transaction ID - Unique ID for each transaction Customer ID - Unique ID of the customer Category - Product category Item - Product name Price Per Unit - Price of one unit Quantity - Number of units purchased Total Spent - Total amount spent on the transaction Payment Method - How the customer paid Location - Online or In-store Transaction Date - Date of the transaction Discount Applied - Whether a discount was applied
Target Column	Category - Used to analyse sales distribution and inventory patterns across product categories
Known Issues	- 1,213 missing Item names - 609 missing Price Per Unit values - 604 missing Quantity values - 604 missing Total Spent values - 4,199 missing Discount Applied values - Transaction Date stored as plain text instead of date format - Quantity stored as decimal instead of whole number - Discount Applied stored as text instead of True/False
Backup Dataset	Online Retail II Dataset Source: UCI Machine Learning Repository https://archive.ics.uci.edu/dataset/502/online+retail+ii

Section 3: Pipeline Design and Decisions

The entire data cleaning process was completed in a single Google Colab notebook. Each stage of the pipeline was organised into its own section, with a brief explanation describing its purpose followed by the corresponding code and output. This structure makes the workflow easy to follow from start to finish. Although the notebook contains 18 numbered steps, the pipeline decisions below have been grouped into 14 stages for clarity.

Step 1: Load the Dataset

I began by loading the dataset using pd.read_csv(). After loading it, I checked the dataset's dimensions, column names, and data types to understand its overall structure before performing any cleaning.

Step 2: Inspect the Data

Before making any changes, I assessed the quality of the dataset using df.info(), df.describe(), df.isnull().sum(), and df.duplicated().sum(). I also recorded a "before" snapshot containing the row count, missing values, and duplicate records. This provided a baseline for comparing the dataset before and after preprocessing.

Step 3: Remove Missing Item Names

The Item column identifies the product involved in each transaction. Records without a product name cannot be used for sales or inventory analysis because the product is unknown. For this reason, I removed all 1,213 rows where the item name was missing, reducing the dataset to 11,362 rows.

Step 4: Check Missing Quantity and Total Spent Values

My initial plan was to remove rows where both Quantity and Total Spent were missing. However, after removing rows with missing item names, I found that no such rows remained. These missing values had already been removed because they belonged to the same records deleted in the previous step.

Step 5: Fill Missing Price Per Unit Values

The Price Per Unit column contained 609 missing values. Instead of deleting those transactions, I filled the missing values with the median price for each product category. I chose the median because it is less affected by unusually high or low prices than the mean. Calculating the median within each category also produced more realistic estimates than using a single value for the entire dataset.

Step 6: Fill Missing Discount Values

The Discount Applied column contained 4,199 missing values. Since a missing entry most likely indicates that no discount was applied, I replaced all missing values with False. This preserved every remaining record while making the column complete and consistent.

Step 7: Convert the Transaction Date

The Transaction Date column was stored as plain text, making date-based analysis difficult. I converted it to the datetime64 format using pd.to_datetime(), allowing it to be handled correctly as a date field.

Step 8: Convert Quantity to Whole Numbers

Although Quantity represents the number of items sold, it was stored as decimal values such as 5.0. Since items are counted as whole units, I converted the column from float64 to int64 using astype(int).

Step 9: Standardize Text Columns

The Category and Item columns contained inconsistent formatting, including extra spaces and inconsistent capitalisation. I removed leading and trailing whitespace using str.strip() and standardized the text using str.title() so that identical values were represented consistently.

Step 10: Encode Categorical Data

To prepare the dataset for future analysis, I encoded the Category, Payment Method, and Location columns using LabelEncoder. The encoded values were added as new columns, while the original text columns were retained so the dataset remained both machine readable and easy to interpret.

Step 11: Normalize Numerical Values

The numerical columns Price Per Unit, Quantity, and Total Spent had different value ranges. To place them on a common scale, I applied MinMaxScaler, which transformed each column to a range between 0 and 1. The original values were preserved, while the normalized values were stored in new columns with the _scaled suffix.

Step 12: Generate the Before and After Data Quality Report

After completing all preprocessing steps, I recorded an "after" snapshot and compared it with the original dataset. The comparison confirmed that all missing values had been handled, the data types had been corrected, and the cleaned dataset contained 11,362 records with no remaining null values.

Step 13: Perform Simple Analysis

With the cleaned dataset ready, I created three bar charts to explore the data. The first chart shows the top five product categories by total sales, the second compares total sales between Online and In store transactions, and the third summarizes sales by payment method. These visualizations provide a quick overview of the cleaned dataset.

Step 14: Export the Final Dataset

Finally, I exported the cleaned dataset as clean_retail_store_sales.csv using df.to_csv(index=False). The exported file contains 11,362 rows and 17 columns, with cleaned, standardized, encoded, and normalized data ready for future analysis.

The pipeline was built using Python 3.10 with Pandas for data loading and cleaning, NumPy for numeric operations, scikit-learn for label encoding and normalization, and Matplotlib and Seaborn for visualization. All code was developed and executed in Google Colab.

Section 4: Before/After Data Quality Report

The table below compares the state of the dataset before any cleaning was applied against the final state after all pipeline steps were completed. All values are taken directly from the notebook cell outputs.

Metric	Before	After
Row Count	12,575	11,362
Rows Removed	-	1,213
Duplicate Rows	0	0
Missing Item	1,213	0
Missing Price Per Unit	609	0
Missing Quantity	604	0
Missing Total Spent	604	0
Missing Discount applied	4,199	0
Transaction Date Type	object (text)	datetime64
Quantity Type	float64	int 64
Discount Applied Type	object (text)	bool
Total Columns	11	17

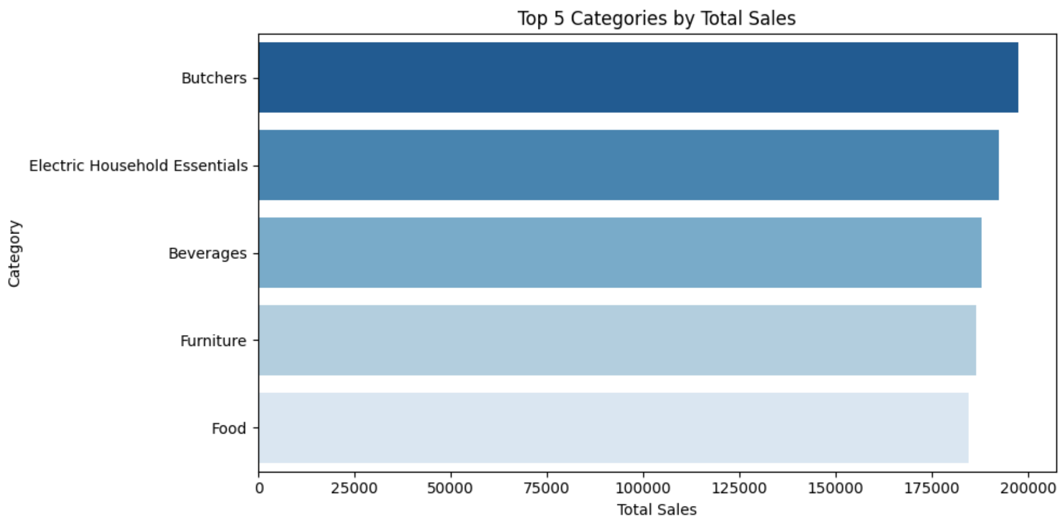
After cleaning, the dataset contains zero missing values across all columns. The only rows removed were the 1,213 records with no item name, which could not be recovered. All other missing values were filled using either the category median or a default False value. Three data type corrections were made and six new columns were added during encoding and normalization, bringing the total column count from 11 to 17.

Section 5: Key Findings

The following findings are based on the original cleaned dataset of 11,362 rows. All figures are derived from grouped aggregations and bar chart outputs produced in the notebook.

Finding 1: Sales by Product Category

Butchers was the top performing category with approximately 200,000 in total sales, followed closely by Electric Household Essentials at 195,000 and Beverages at 190,000. Furniture and Food rounded out the top five at approximately 188,000 and 185,000 respectively. The narrow gap between all five categories suggests broadly balanced demand across product types, with no single category dominating revenue.

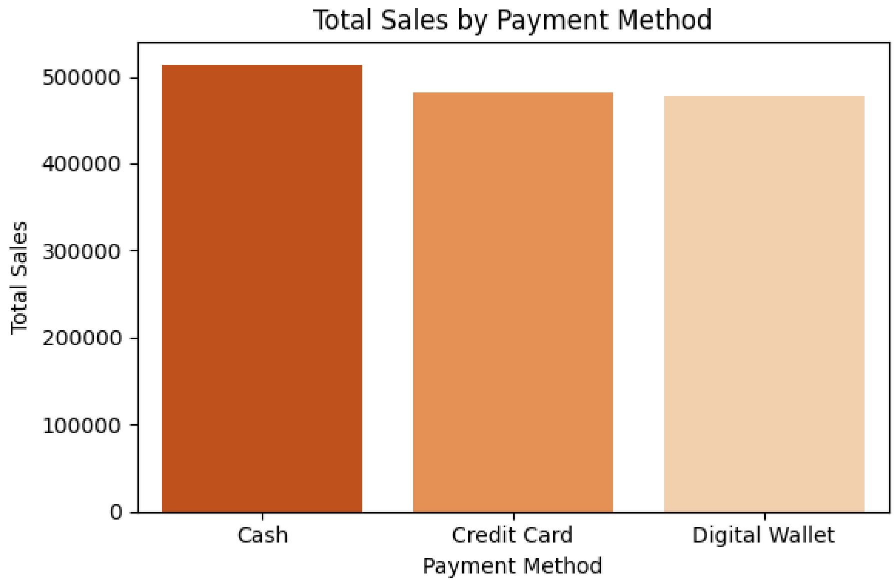


Finding 2: Sales by Location

Online transactions generated approximately 730,000 in total sales compared to 715,000 for In-store transactions, a difference of around 15,000. The two channels contributed almost equally to overall revenue, indicating that the business serves both its online and physical customer bases effectively. Neither channel shows a significant advantage, which suggests the business does not rely heavily on one sales channel over the other.

Finding 3: Sales by Payment Method

Cash was the most used payment method with approximately 510,000 in total sales, followed by Credit Card at 480,000 and Digital Wallet at 470,000. The three methods are relatively close in total volume, suggesting customers use a mix of payment options rather than one dominant method. This spread may have implications for payment infrastructure planning, as all three methods need to be well supported.



Section 6: Limitations and Conclusion

Limitations

This project used a retail dataset sourced from Kaggle that was intentionally created with data quality issues for cleaning practice. It does not represent real transactions from an actual business, so the results presented in the analysis should not be interpreted as genuine retail insights. The patterns observed in product categories, payment methods, and sales distribution are likely influenced by the way the dataset was created rather than real customer behaviour.

The method used to fill missing Price Per Unit values also has limitations. Missing values were replaced using the median price within each product category. While this is a practical approach, it assumes that products within the same category have similar prices. If a category contains both low-priced and high-priced products, the imputed value may not accurately reflect the original missing price.

The analysis itself is fairly simple. Only three bar charts were created to summarize sales by product category, location, and payment method. The project does not explore other useful questions, such as sales trends over time, customer purchasing behaviour, or the relationship between discounts and payment methods. A more detailed analysis would require additional visualizations and statistical techniques.

Conclusion

This project developed a complete data cleaning pipeline for a retail transactions dataset sourced from Kaggle. The original dataset contained 12,575 rows and 11 columns, with 7,229 missing values across five columns and several incorrect data types. After cleaning, the final dataset contained 11,362 records with no missing values and correctly formatted data types.

In addition to cleaning the data, six new columns were created through categorical encoding and feature normalization to prepare the dataset for future analysis. Finally, three bar charts were generated from the cleaned dataset, providing a basic overview of sales across product categories, shopping locations, and payment methods. Although the dataset is not based on real business transactions, the project demonstrates a complete and well-documented data preprocessing workflow, producing a clean dataset that is ready for further analysis.

Project Links

Google Colab: Google Colab
GitHub Repo: GitHub
